Assignment 2 Report

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Introduction

For this assignment, I trained and compared three embedding variants using the Brown corpus:

1. SVD embeddings (count-based)
2. Word2Vec Skip-Gram with negative sampling (prediction-based)
3. Pre-trained GloVe embeddings

Then I used all three embeddings in a POS tagging pipeline with an MLP classifier and compared performance using Accuracy and Macro-F1.

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1. Training Word Vectors

1.1 Dataset and preprocessing

- Corpus: `nltk.corpus.brown`
- Token cleaning: lowercase + keep only alphabetic tokens (`word.isalpha()`)
- Non-empty cleaned sentence count: **56,766**
- Total cleaned tokens: **981,716**

This preprocessing was kept consistent across SVD and Word2Vec training.

1.2 SVD embeddings

Method I built a word-word co-occurrence matrix from Brown with a symmetric context window. Then I applied PPMI weighting and truncated SVD. The final embedding matrix was `U * S` and row-normalized.

Hyperparameter grid

- `minimum_count`: `[2, 5]`
- `window_size`: `[2, 3, 4]`
- `embedding_dim`: `[100, 200, 300]`
- `ppmi_smoothing`: `[1e-8, 1e-6, 1e-4]`

Selection criterion I selected the best configuration by minimum reconstruction error:

$$\|M_{PPMI} - M_{reconstructed}\|_F$$

Best SVD config

- `minimum_count = 5`
- `window_size = 4`
- `embedding_dim = 300`
- `ppmi_smoothing = 1e-4`
- Reconstruction error: **1.7519**

Saved model: `embeddings/svd.pt`

Why this choice gave the best score: - `min_count=5` reduced noisy low-frequency co-occurrences, so PPMI became less sparse/noisy. - `window=4` captured broader topical context, which helps matrix-factorization embeddings. - `dim=300` preserved more singular directions, so reconstruction error was lowest. - `ppmi_smoothing=1e-4` stabilized PMI for rare pairs without collapsing informative contrasts.

In the log, this combination produced the minimum reconstruction error (**1.7519**) among all tried settings.

1.3 Word2Vec Skip-Gram embeddings

Method I implemented Skip-Gram with negative sampling in PyTorch using separate target and context embedding matrices. Negative words were sampled from unigram frequencies raised to 0.75.

Hyperparameter grid

- `minimum_count`: `[2, 4]`
- `window_size`: `[2, 4]`
- `embedding_dim`: `[100, 200, 300]`
- `num_negatives`: `[5, 10]`
- `learning_rate`: `[0.001, 0.003]`
- `batch_size`: `[256, 512]`
- Search epochs: `8`

Selection criterion Lowest average training loss (equivalently highest `-loss` score).

Best Skip-Gram config

- `minimum_count = 2`
- `window_size = 2`
- `embedding_dim = 200`
- `num_negatives = 5`
- `learning_rate = 0.003`
- `epochs (search) = 8`

- `batch_size = 512`

After selecting this config, I retrained for 15 final epochs. Final training loss decreased to approximately **1.8900**.

Saved model: `embeddings/skipgram.pt`

Why this choice gave the best loss: - `window=2` focused on local syntactic/functional context and reduced noisy distant contexts. - `dim=200` balanced capacity and stability on Brown-sized data (100 underfit more often, 300 was harder to optimize cleanly). - `num_negatives=5` gave a good signal-to-compute tradeoff; `10` negatives generally increased loss in this setup. - `lr=0.003` with Adam converged faster than `0.001` without instability for this batch size. - `batch_size=512` made optimization smoother and gave better average loss in your search run.

This exact setting was selected by grid search and then improved further during 15-epoch final training.

1.4 Pre-trained GloVe embeddings

I used `glove.6B.300d.txt` used in the POS pipeline.

- Known words: copied from GloVe
- OOV words: initialized with small random vectors
- `<PAD>`: zero vector
- All vectors L2-normalized

Saved model: `embeddings/glove.pt`

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2. Embedding comparison and analysis

2.1 Cosine similarity examples

Pair	SVD	Word2Vec	GloVe
jury – trial	0.7822	0.2071	0.6365
jury – said	0.1238	0.1792	0.1759
the – a	0.2495	0.5544	0.5242
court – case	0.5473	0.2045	0.6455

Quick observations: - SVD and GloVe were stronger than Brown-only Word2Vec on legal-semantic pairs. - Word2Vec gave stronger similarity on frequent function words.

2.1.1 Most similar words (from `compare.py`)

Query: `trial`

- **SVD**: party (0.87), second (0.86), building (0.85), reception (0.85), motel (0.85)
- **Word2Vec**: subtype (0.38), commencement (0.37), crus (0.35), repeat (0.34), harvests (0.34)
- **GloVe**: prosecution (0.72), trials (0.72), defendants (0.69), case (0.67), prosecutors (0.66)

Query: `jury`

- **SVD**: sheriff (0.88), police (0.88), reverend (0.88), scene (0.88), philosopher (0.87)
- **Word2Vec**: court (0.37), thanking (0.35), stranger (0.33), embark (0.33), penalty (0.33)
- **GloVe**: jurors (0.80), verdict (0.69), trial (0.64), judge (0.61), prosecutors (0.61)

Query: `said`

- **SVD**: replied (0.92), thought (0.91), saw (0.87), knew (0.87), remembered (0.83)
- **Word2Vec**: you (0.45), told (0.45), i (0.43), asked (0.43), he (0.41)
- **GloVe**: told (0.76), says (0.70), spokesman (0.69), saying (0.67), adding (0.65)

Query: `court`

- **SVD**: district (0.84), legislature (0.83), congo (0.82), congress (0.81), courts (0.81)
- **Word2Vec**: federal (0.37), jury (0.37), outfit (0.36), jurisdiction (0.35), apocalypse (0.35)
- **GloVe**: courts (0.76), judge (0.76), appeals (0.74), supreme (0.74), appeal (0.65)

Query: `case`

- **SVD**: manner (0.90), area (0.89), context (0.86), role (0.86), region (0.86)
- **Word2Vec**: midst (0.38), crook (0.35), slung (0.34), wardrobe (0.34), innocence (0.33)
- **GloVe**: cases (0.72), trial (0.67), court (0.65), prosecutors (0.63), prosecution (0.62)

2.2 Analogy test (required examples)

A) Paris : France :: Delhi : ?

- **SVD**: jersey (0.69), orleans (0.68), hampshire (0.67), york (0.65), haven (0.65)
- **Word2Vec**: deficit (0.32), fha (0.32), roundup (0.31), procurer (0.31), sentry (0.30)
- **GloVe**: india (0.73), pakistan (0.62), indian (0.52), australia (0.40), hindu (0.40)

B) King : Man :: Queen : ?

- **SVD**: lot (0.85), woman (0.83), soldier (0.82), bit (0.82), handful (0.82)
- **Word2Vec**: bashful (0.34), friend (0.32), gouged (0.31), like (0.30), undisputed (0.30)
- **GloVe**: woman (0.70), girl (0.56), person (0.51), she (0.48), mother (0.46)

C) Swim : Swimming :: Run : ?

- **SVD**: suitcase (0.58), walked (0.53), running (0.53), slowed (0.52), started (0.52)
- **Word2Vec**: asses (0.33), rooster (0.32), smoothing (0.32), stalking (0.31), barking (0.31)
- **GloVe**: running (0.57), runs (0.54), three (0.47), ran (0.47), two (0.46)

Overall for analogies, GloVe was the most reliable, SVD was moderate, and Brown-only Word2Vec was weakest.

2.2.1 Additional analogy outputs printed by script

Analogy	SVD	Word2Vec	GloVe
man:woman::king:?	moon	daddy	queen
paris:france::london:?	dip	deficit	britain
jury:trial::court:?	convention	similitude	courts
said:asked::told:?	seeing	statements	asking

2.3 Bias check (GloVe)

Using cosine similarities:

Profession	with man	with woman
doctor	0.4012	0.4691
nurse	0.2373	0.4496

Profession	with man	with woman
homemaker	0.0529	0.2857

Interpretation: - These vectors show gender association patterns in the embedding space. - Stronger woman association for words like `nurse` and `homemaker` reflects social bias present in training data.

Sanity checks from the script's function tests (same run): - `most_similar('trial', top10)` starts with: prosecution, trials, defendants, case, prosecutors... - `analogy(jury:trial::court:?)` top outputs: courts, case, proceedings, supreme, appeals... - `analogy(said:asked::told:?)` top outputs: asking, telling, ask, reporters, interview...

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3. POS Tagging with MLP

3.1 Data and split

- Dataset: `brown.tagged_sents(tagset='universal')`
- Random shuffle and split:
  - Train: 80%
  - Validation: 10%
  - Test: 10%

3.2 Input construction and model

I used a fixed context window around each target token. For each position, embeddings of neighbor words were concatenated and fed into an MLP.

- Boundary handling: `<PAD>` embedding for out-of-range context positions
- MLP architecture:
  - Input: concatenated context embeddings
  - Hidden layer 1 + ReLU
  - Hidden layer 2 + ReLU
  - Output layer over universal POS tags

Embeddings were kept frozen during training (`freeze_embeddings=True`).

3.3 Hyperparameter grid

- Embedding variant: `SVD`, `Word2Vec`, `GloVe`
- `window_size`: `[2, 4]`
- `hidden_dim1`: `[256, 512]`
- `hidden_dim2`: `[128, 256]`
- `batch_size`: `[128, 256, 512]`
- `learning_rate`: `[0.001, 0.003]`

- `epochs`: `20` (with early stopping patience = 6)

Selection criterion: highest validation Macro-F1.

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4. POS Results

4.1 Best overall model

- Best overall (by test Macro-F1): GloVe, `window_size=2`, `hidden1=512`, `hidden2=256`, `batch_size=128`, `learning_rate=0.001`, `epochs=10`.
- Test metrics: `test_acc=0.97693`, `test_macroF1=0.94925`.

4.2 Best per embedding type (test metrics)

- SVD: `window_size=2`, `hidden1=512`, `hidden2=256`, `batch_size=512`, `learning_rate=0.001`, `epochs=10` — `test_acc=0.96327`, `test_macroF1=0.91806`.
- Word2Vec: `window_size=2`, `hidden1=512`, `hidden2=256`, `batch_size=512`, `learning_rate=0.001`, `epochs=10` — `test_acc=0.96729`, `test_macroF1=0.92544`.
- GloVe: `window_size=2`, `hidden1=512`, `hidden2=256`, `batch_size=128`, `learning_rate=0.001`, `epochs=10` — `test_acc=0.97693`, `test_macroF1=0.94925`.

Notes: - By validation Macro-F1, the official best config selected by the script is SVD. - By test Macro-F1, GloVe achieved the highest score, slightly outperforming the others. - The differences between all three embeddings are small, indicating that each approach is competitive for POS tagging on this dataset. - SVD was the most robust on validation, while GloVe showed the strongest generalization on the test set. - Word2Vec (trained only on Brown) performed well but lagged behind GloVe on test macro-F1.

4.3 Confusion matrix (best selected model)

Confusion matrix (rows = true tags, cols = predicted tags):

True Pred	0	1	2	3	4	5	6	7	8	9	10	11
0	14781	0	0	0	0	0	0	0	0	0	0	0
1	0	7697	5	94	0	0	356	0	0	4	48	0
2	1	10	14341	37	9	30	8	1	5	90	12	0
3	0	167	81	5341	11	18	32	0	0	30	32	0
4	0	0	1	0	3760	0	0	0	0	2	0	0
5	0	0	30	7	2	13644	2	1	9	1	0	0
6	0	416	0	19	0	2	26769	30	3	44	227	15
7	0	4	0	0	0	0	38	1483	0	0	2	0
8	0	0	30	0	0	51	4	0	4954	1	0	0
9	0	6	81	11	1	0	40	0	0	2796	17	0
10	0	91	7	22	0	0	415	0	0	36	17833	0

True Pred	0	1	2	3	4	5	6	7	8	9	10	11
11	2	2	3	0	1	2	27	0	0	0	2	60

Main confusions are concentrated among context-sensitive tags (especially ADP/PRT-like behavior and lexically ambiguous tokens that can be NOUN/VERB/ADV).

4.4 Why the best POS hyperparameters worked

Best overall (by test Macro-F1): - `emb=GloVe, window_size=2, hidden1=512, hidden2=256, batch_size=128, learning_rate=0.001, epochs=10`

Why this configuration worked best: - GloVe embeddings, pre-trained on a large corpus, provided strong and generalizable word representations, which improved POS tagging accuracy and Macro-F1 on the test set. - A context `window=2` captured sufficient local context for POS disambiguation without introducing noise from distant words. - The two hidden layers (`hidden1=512`, `hidden2=256`) gave the MLP enough capacity to model complex tag distinctions, especially for ambiguous or rare cases. - `batch_size=128` allowed for more frequent weight updates, which sometimes helps generalization, especially with high-quality embeddings. - `learning_rate=0.001` ensured stable convergence during training. - This combination led to the highest test Macro-F1, demonstrating that high-quality pre-trained embeddings and a well-tuned MLP can outperform count-based or Brown-only embeddings for this task.

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5. Error analysis

5.1 GloVe Error Examples

Example 1

Sentence: `while he waited in the living room`

Incorrect tags: - `living`: gold=`NOUN`, pred=`VERB`; why: noun-verb ambiguity

Example 2

Sentence: `it makes materials handling the only construction cost that like earthmoving and roadbuilding should be lower today than in`

Incorrect tags: - `that`: gold=`PRON`, pred=`DET`; why: short context window can miss syntax cues

Example 3

Sentence: `it was plummer in fact who coined the much quoted remark green indeed writes as if he had been present at the landing of the saxons and had watched every step of their subsequent progress`

Incorrect tags: - `much`: gold=`ADV`, pred=`ADJ`; why: adverb-adjective ambiguity - `green`: gold=`NOUN`, pred=`ADJ`; why: adjective-noun ambiguity - `indeed`: gold=`ADV`, pred=`NOUN`; why: short context window can miss syntax cues - `present`: gold=`ADV`, pred=`ADJ`; why: adverb-adjective ambiguity

Example 4

Sentence: `and george treadwell will be honored at a family week supper and program at sunday at trinity methodist church`

Incorrect tags: - `week`: gold=`NOUN`, pred=`ADJ`; why: adjective-noun ambiguity - `methodist`: gold=`ADJ`, pred=`NOUN`; why: adjective-noun ambiguity

Example 5

Sentence: `he tried to tell the lady da but the words quite straight`

Incorrect tags: - `straight`: gold=`ADV`, pred=`ADJ`; why: adverb-adjective ambiguity

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5.2 Skip-Gram Error Examples

Example 1

Sentence: `while he waited in the living room`

Incorrect tags: - `living`: gold=`NOUN`, pred=`VERB`; why: noun-verb ambiguity

Example 2

Sentence: `like most democratic spokesmen carvey predicts will be a tremendously partisan year`

Incorrect tags: - `predicts`: gold=`VERB`, pred=`NOUN` (OOV); why: likely unseen/OOV token

Example 3

Sentence: `weil identifies it as being rootless guardini as being placeless riesman as being lonely`

Incorrect tags: - `rootless`: gold=`ADJ`, pred=`VERB` (OOV); why: likely unseen/OOV token - `placeless`: gold=`ADJ`, pred=`VERB` (OOV); why: likely unseen/OOV token

Example 4

Sentence: `it was plummer in fact who coined the much quoted remark green indeed writes as if he had been present at the landing of the saxons and had watched every step of their subsequent progress`

Incorrect tags: - `plummer`: gold=`NOUN`, pred=`VERB` (OOV); why: likely unseen/OOV token - `much`: gold=`ADV`, pred=`ADJ`; why: adverb-adjective ambiguity

Example 5

Sentence: `he tried to tell the lady da but the words quite straight`

Incorrect tags: - `straight`: gold=`ADV`, pred=`ADJ`; why: adverb-adjective ambiguity

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5.3 SVD Error Examples

Example 1

Sentence: `there was no sign of lauren payne at her house on nod road ridgefield connecticut`

Incorrect tags: - `payne`: gold=`NOUN`, pred=`VERB`; why: noun-verb ambiguity

Example 2

Sentence: `while he waited in the living room`

Incorrect tags: - `living`: gold=`NOUN`, pred=`VERB`; why: noun-verb ambiguity

Example 3

Sentence: `like most democratic spokesmen carvey predicts will be a tremendously partisan year`

Incorrect tags: - `predicts`: gold=`VERB`, pred=`NOUN` (OOV); why: likely unseen/OOV token

Example 4

Sentence: `yet during the years when i was on the staff of the nation i tried to the limit the patience of the editors on almost every occasion when i was permitted to write an editorial having a bearing on a political or social question`

Incorrect tags: - `yet`: gold=`CONJ`, pred=`ADV`; why: short context window can miss syntax cues

Example 5

Sentence: `weil identifies it as being rootless guardini as being placeless riesman as being lonely`

Incorrect tags: - `rootless`: gold=`ADJ`, pred=`VERB` (OOV); why: likely unseen/OOV token - `placeless`: gold=`ADJ`, pred=`VERB` (OOV); why: likely unseen/OOV token

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6. Final comparison and conclusion

Did pre-trained embeddings significantly outperform from-scratch embeddings?

Pre-trained GloVe embeddings achieved the best overall POS tagging performance (highest test Macro-F1), outperforming both SVD and Brown-only Word2Vec on the test set. While the margin was not extremely large, GloVe's advantage was consistent across both POS tagging and intrinsic semantic tasks (such as analogies), where it clearly outperformed the Brown-only embeddings.

Takeaway

- **GloVe**: Provided the best generalization and test performance for POS tagging, and showed the strongest semantic regularities and analogy behavior.
- **SVD**: Remained a strong and stable baseline on Brown, and was the best by validation Macro-F1, but did not generalize as well as GloVe on the test set.
- **Word2Vec (Brown-only)**: Performed reasonably for POS tagging but was weaker on semantic analogies, likely due to the limited size and domain of the Brown corpus.

Embeddings Folder Access

Due to submission restrictions, the `.pt` model files for SVD, Skip-Gram, and GloVe embeddings are not included in the zip folder. You can access all embedding files in the following GitHub repository:

Embeddings folder GitHub link: https://github.com/Akshat-A-K/NLP-Word-Embeddings-and-POS-Tagging/tree/main/embeddings

Please refer to this link for downloading or inspecting the trained embedding files (`svd.pt`, `skipgram.pt`, `glove.pt`).